Mayesh Mohapatra

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EDUCATION

Texas A&M University — Aug. 2025 – May 2027
Master of Science in Computer Science GPA: 3.67 / 4.0 — *College Station, TX*

- **Relevant Coursework:** Deep Learning, Computer Vision and Robotic Perception, Deep Reinforcement Learning, ML for 3D Vision and Graphics, Generative AI for CV, Advanced CV

Bennett University — May 2023
B.Tech (Honors), Computer Science Engineering GPA: 3.54 / 4.0 — *Greater Noida, India*

- **Relevant Coursework:** Machine Learning, Data Structures and Algorithms, Database Systems, Linear Algebra, Probability and Statistics, Computer Networks

TECHNICAL SKILLS

Programming: Pythonx, CUDA, C++
ML Frameworks: PyTorch,Hugging Face Diffusers / Transformers, Scikit-Learn, NumPy, Weights and Biases
Generative Models: Diffusion Transformers (DiT, FLUX.1, SD3), Flow Matching, Rectified Flow, DDPM/DDIM, Consistency Models, Classifier-Free Guidance, Score-Based Generative Models, Variational Autoencoders
Vision Backbones: DINOv2 (ViT-B/14), Vision Transformers, CLIP
Computer Vision: 3D Scene Reconstruction, Multi-View Geometry, Human Pose Estimation, Semantic Segmentation, Point Cloud Processing, Multi-Modal Fusion
Systems & Tools: Git, Docker, CUDA Toolkit, PyTorch Profiler, Weights & Biases, Linux, JupyterLab

EXPERIENCE

Texas A&M University, Graduate Researcher (3D Perception) — Aug. 2025 – Present
Advisor: Prof. Cheng Zhang — *College Station, TX*

- Built an end-to-end monocular 3D scene reconstruction pipeline for dynamic athlete reconstruction from single-view RGB video, integrating Human3R and VGGT for pose estimation with DeepLabv3 segmentation into a unified PyTorch tracking framework; validated on internal multi-athlete benchmarks.
- Cut end-to-end inference latency by 40% for real-time biomechanical feedback during live training sessions, using mixed-precision CUDA kernels and INT8 quantization across the full perception stack.
- Built an analytics module that extracts biomechanical metrics (jump height, center-of-mass velocity, take-off angle) across 3+ athletic disciplines by fusing 3D pose outputs with physics-based motion analysis.
- Delivered production perception APIs consumed by biomechanics researchers and coaching staff, with standardized data contracts linking 3D reconstruction outputs to a downstream decision-support system.

Indian Institute of Science (IISc) & ISRO, Project Associate — May 2023 – May 2025
Multi-Modal Perception and Satellite Image Analysis — *Bengaluru, India*

- Boosted model training throughput by 30% on 100+ GB/day of HDF5 multispectral satellite imagery by building an automated Python ingestion pipeline that generated roughly 14,000 annotated frames via GDAL and Rasterio.
- Designed a two-stage cyclone detector (Center-Locator and Intensity-Estimator) in PyTorch with self-supervised pre-training on unlabeled satellite data, validated across 5+ spectral sensor bands using mAP and RMSE.
- Reduced inference latency from 3-4 hours to under 1 second (10,000x speedup) for near-real-time cyclone monitoring, applying model pruning, INT8 quantization, and parallelized preprocessing.
- Scaled training to 200+ tracked experiments with reproducible deployments using Weights & Biases, Docker, and distributed Spark/Dask data workflows.
- Published research at **IEEE IGARSS 2025** (Brisbane) on deep-learning models for cyclone detection on infrared satellite imagery, covering architecture design, ablations, and deployment results.

Georgia Institute of Technology, Research Intern (Remote) — Jan. 2023 – May 2023
Generative Models and Anomaly Detection — *Atlanta, GA*

- Improved out-of-distribution robustness by 15% on insider-threat detection by synthesizing rare-class training data with GANs in PyTorch to expand minority-class coverage.
- Achieved 95% classification accuracy and lifted rare-event recall by 12%, training autoencoder-based anomaly detectors combined with a Random Forest and Logistic Regression ensemble.
- Cut data preprocessing time by 25% across 5+ dataset splits by vectorizing cleaning workflows in Pandas and NumPy.

Deloitte Haskins & Sells LLP, Data Science Intern — Jun. 2022 – Mar. 2023
ML Pipelines and Analytics — *Bengaluru, India*

- Saved 15 engineering hours/week and cut audit review time by 40% by refactoring Spark ETL pipelines and overhauling journal-entry testing logic across distributed data clusters.
- Deployed analytics dashboards adopted by 4+ audit teams (85% deployment rate) by building Power BI exception-analysis tools and automating reporting workflows.

SELECTED PROJECTS

Hierarchically Decomposed Diffusion Guidance (HDG) *Manuscript in Progress* Feb. 2026 – Present

- Co-developed HDG, a method that decomposes classifier-free guidance into a sequence of hierarchical components for fine-grained conditional diffusion generation, improving stability, sample quality, and controllability over standard CFG on hierarchically-labeled benchmarks.
- Built data pipelines that align fine-grained class hierarchies with diffusion conditioning signals across multiple semantic taxonomies.
- Implemented the full evaluation suite measuring generation quality, controllability, and fine-grained alignment with ablation studies isolating each decomposition component against standard CFG baselines.

Pose4DGS: Pose-Conditioned 4D Gaussian Splatting | *PyTorch, CUDA, DINOv2, SMPL, W&B* Jan. 2026 – May 2026

- Improved unseen-pose generalization on monocular 4D Gaussian Splatting human avatars by introducing two zero-overhead training-only modules: SMPL-based Pose-Space Augmentation and DINOv2 (ViT-B/14) feature supervision.
- Designed a 3-tier novel-pose evaluation protocol (easy / moderate / extreme) on ZJU-MoCap and PeopleSnapshot, running a full 2x2 ablation grid and compressing DINOv2 embeddings 768 to 64 via PCA for semantic body-part consistency.
- Led a 3-person team across rendering, deformation, and evaluation modules, tracking all ablations in W&B and delivering a reproducible 6-8 page technical report.

Monocular 3D Human Reconstruction | *PyTorch, CUDA, NeRF, Gaussian Splatting, SMPL, COLMAP* Aug. 2025 – Present

- Building a neural rendering pipeline that converts single-view RGB video into pose-driven 3D human meshes using neural implicit representations and SMPL priors, targeting sub-centimeter surface accuracy on ZJU-MoCap and Human3.6M.
- Benchmarked reconstruction quality against state-of-the-art baselines on PSNR, SSIM, and LPIPS by iterating on deformation-network architectures and loss functions to improve cross-subject generalization.

Recurrent Attention Model (RAM) | *PyTorch, REINFORCE* Aug. 2025 – Dec. 2025

- Achieved 96.64% MNIST test accuracy using only 6 glimpses (roughly 49% of input pixels), matching full-image CNN baselines at a fraction of the compute, by reproducing RAM in PyTorch with REINFORCE policy gradients.
- Stabilized policy-gradient training and reduced variance with a learned value baseline; ablated glimpse count (4/6/8), patch resolution, and reward shaping; released a reproducible open-source codebase.

CycloAI: Multi-Modal Cyclone Detection Pipeline | *PyTorch, GDAL, Rasterio, Spark, Dask, Docker* May 2023 – May 2025

- Reduced inference from 3-4 hours to under 1 second on 100+ GB/day satellite streams by co-building a two-stage detector with self-supervised pre-training, pruning, and INT8 quantization; published at IEEE IGARSS 2025.
- Lifted minority-class recall by 12% across 5+ spectral bands using focal loss and SMOTE, tracking 200+ runs in W&B inside Docker-containerized environments.

PUBLICATIONS

- **Comparative Analysis of Deep Learning Models for Cyclone Detection and Localisation on Infrared Satellite Images.** A. Agrawal*, **M. Mohapatra***, A. Raja, P. Tiwari, V. Pattanaik, A. Agarwal, N. Jaiswal, P. Rathore. *IEEE IGARSS 2025*, Brisbane, Australia.
- **A Self-Supervised Visual Analytical Approach for Automatic Detection of Cyclonic Events in Satellite Observations.** A. Agrawal, **M. Mohapatra**, P. Tiwari, V. Pattanaik, A. Agarwal, N. Jaiswal, P. Rathore. *40th Annual Symposium on Space Science and Technology*, ISRO-IISc, 2024.
- **GAN-Based Approach for Synthetically Generating Insider Threat Scenarios. M. Mohapatra**, A. Phukan, V. Madisetti. *Journal of Software Engineering and Applications*, 16(11):586–604.